Vaibhav Rathore

Senior AI Data Scientist at MOFSL and Ph.D. Scholar at CSRE, IIT Bombay; M.S. by Research alumnus of C-MInDS, IIT Bombay.

Mumbai, India · vaibhav.rathor.in@gmail.com

Website · Google Scholar · ORCID · GitHub · LinkedIn

Experience

Senior AI Data Scientist Nov 2025 - Present
Motilal Oswal Financial Services Ltd. (MOFSL)
AI Research Team; time-series and tabular foundation models for financial data, India-focused AI, Indic TTS/ASR tokenization, lip-sync systems, and research for top-tier venues. Public details are intentionally high-level because the work is proprietary.

Research Intern, Vehant Fellow Track Sep 2025 - Nov 2025
Vehant Technologies
Pedestrian Attribute Recognition using hyperbolic, graph-based, correlation-aware, and vision-language methods.

Research Intern Sep 2025 - Oct 2025
Motilal Oswal Financial Services Ltd. (MOFSL)
Evaluated lip-sync and Gaussian-Splatting approaches for synchronization and 3D avatars.

Research Intern May 2025 - Jul 2025
Sony Research India
3D lip synchronization using Gaussian Splatting and controllable 3D avatars.

Research & Development Intern May 2024 - Jul 2024
Clinical AI Assistance
LoRA fine-tuning and multimodal medical question answering.

Graduate Engineer Trainee Aug 2022 - Jul 2023
Reliance Industries Ltd.
Predictive maintenance and operational analytics.

Education

Ph.D. · IIT Bombay 2026 - Present
Centre of Studies in Resources Engineering (CSRE) · CPI 9.00

M.S. by Research · C-MInDS, IIT Bombay 2023 - 2025
Artificial Intelligence and Data Science · CPI 9.31

B.Tech. · MNNIT Allahabad 2018 - 2022
Mechanical Engineering · CPI 9.10

Recognition

Award for Excellence in Research 22 Aug 2026
C-MInDS, IIT Bombay. Awarded for M.S. by Research excellence.

GATE 2022
97.84 percentile in Engineering Sciences (XE); 96 percentile in Mechanical Engineering (ME).

JEE Main 2018
Top one percentile.

Teaching & mentoring

DS 303: Introduction to Machine Learning Jul 2024 - Present
Teaching Assistant, IIT Bombay

e-PGD: Introduction to Python Programming Jan 2025 - Present
Teaching Assistant, IIT Bombay

Student mentor · MNNIT Allahabad Jul 2021 - May 2022
Mentored 20 students.

Academic service

AAAI · Program Committee · 2027

AISTATS · Reviewer · 2027

ICVGIP · Reviewer · 2026

NeurIPS Position Paper Track · Reviewer · 2026

International Journal of Computer Vision · Reviewer · 2025-2026

Skills & research interests

Python · PyTorch · Computer Vision · Indic NLP · Gaussian Splatting · Financial Time Series

Domain Generalization · Generalized Category Discovery · Fine-Grained Recognition · Open-World Learning · Representation Learning

Selected coursework

Ph.D. · IIT Bombay
Optimization in Machine Learning · Principles of Satellite Image Processing · Seminar

M.S. by Research · C-MInDS, IIT Bombay
Foundations of Machine Learning · Advanced Topics in Deep Learning · Medical Image Computing · Algorithms and Complexity · Machine Learning for Remote Sensing II · Statistical Signal Analysis · Image Processing · Optimisation

Earlier education

ISC · De Paul School · 2015-2017 · 98.00%

ICSE · De Paul School · 2013-2015 · 93.80%

Conferences & journals

Published and accepted work. Acceptance is not a claim of completed presentation or archival publication.

HElIOS: Hyperbolic Generalized Category Discovery for Remote Sensing

WACV 2027 · Accepted

Sadia Hussain, **Vaibhav Rathore**, Divyam Gupta, Saurabh Prasad, Aleksandra Pizurica, Biplab Banerjee

Vaibhav's affiliation: IIT Bombay

Introduces a hyperbolic framework for discovering known and novel categories in remote-sensing imagery.

SuTRA: Structurally-Unified Tokenization with Root Awareness

Interspeech 2026 · Accepted

Vaibhav Rathore, Siddhant Gole, Dadhichi Telwadkar, Rooshil Bhatia, Maulik Ruparel, Siddharth Surekha, Neha Bhargava

A morphology-aware tokenizer that preserves Indic akshara structure and reduces morphological shattering across Hindi, Marathi, and Gujarati.

Paper · Code · Project

M³ QuestionIng: Multi-modal Multi-span Medical Question Answering

ACM Transactions on Computing for Healthcare · Published

Anisha Saha, **Vaibhav Rathore**, Abhisek Tiwari, Akash Ghosh, Sai Ruthvik Edara, Sriparna Saha

Introduces M³QAFrame and a medical question-answering dataset whose answers draw jointly from multiple textual and visual spans.

Paper · Code & Dataset

HiDISC: A Hyperbolic Framework for Domain Generalization with Generalized Category Discovery

NeurIPS 2025 · Published

Vaibhav Rathore, Divyam Gupta, Biplab Banerjee

A hyperbolic framework for cross-domain alignment and novel-category discovery without expensive episodic simulation.

Paper · Code · Project · Slides · Poster

When Domain Generalization Meets Generalized Category Discovery: An Adaptive Task-Arithmetic Driven Approach

CVPR 2025 · Published

Vaibhav Rathore, Shubhranil B, Saikat Dutta, Sarthak Mehrotra, Zsolt Kira, Biplab Banerjee

Introduces DG²CD-Net and the DG-GCD setting, combining episodic adaptation with adaptive task arithmetic.

Paper · Code · Project · Presentation · Poster

Workshops & posters

Published and accepted work. Acceptance is not a claim of completed presentation or archival publication.

Align Before You Quantize: Making Leech Lattice Tokenization Work for Time Series

AI4F Workshop at ACM ICAIF 2026 · Accepted poster · Non-archival workshop

Akanksh Adugani Vanjari, **Vaibhav Rathore**, Siddhant Gole, Siddharth Sureka, Neha Bhargava

Vaibhav's affiliation: Motilal Oswal Financial Services

Uses hyperspherical alignment and phased training to improve lattice-codebook utilization and reconstruction for time-series tokenization.

CandleDiff: Contextual Diffusion for Structurally Valid OHLC Generation

AI4F Workshop at ACM ICAIF 2026 · Accepted poster · Non-archival workshop

Siddhant Gole, Nachiketa Ratnakar Patil, **Vaibhav Rathore**, Siddharth Sureka, Neha Bhargava

Vaibhav's affiliation: Motilal Oswal Financial Services

Introduces contextual diffusion for controllable firm-level OHLC generation with guaranteed candle geometry and market-aware conditioning.

Position: Many Agents, Few Minds - Model Concentration as a Governance Metric for Agent-Native Financial Markets

AI4F Workshop at ACM ICAIF 2026 · Accepted poster · Non-archival workshop

Vaibhav Rathore, Siddhant Gole, Siddharth Sureka, Neha Bhargava

Vaibhav's affiliation: Motilal Oswal Financial Services

Argues that agent-based finance can decentralize intermediation while concentrating decisions in shared foundation models, and proposes a metric for that risk.

Beyond Accuracy: What Language-Subspace Interventions Capture, Remove, and Improve

Multi-LLL Workshop at AACL-IJCNLP 2026 · Accepted

Shaanil Punglia, Dadhichi Telwadkar, **Vaibhav Rathore**, Siddharth Sureka, Neha Bhargava

Evaluates whether language-subspace interventions truly remove language information and improve multilingual mathematical reasoning without sacrificing response-language fidelity.

Towards Unconstrained Pedestrian Attribute Recognition: Evaluating Domain Shift and Open-World Constraints

NewInML Workshop at NeurIPS 2026 · Poster

Dattaraj Salunkhe, **Vaibhav Rathore**, Biplab Banerjee

Studies pedestrian attribute recognition under domain shift and open-world constraints.

Ongoing research is listed separately on the website; submission and review details are not disclosed.